POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5024D | Semester V | Theory | 4 Credits

Advanced Production Process

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5024D as Advanced Production Process in Semester V for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user-approved public-source approach, this handbook is transparently based on public diploma lecture notes from Vedang Institute of Technology and the IIT Guwahati NPTEL Advanced Machining Processes course outline. It is a learning handbook and does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to the neighbouring handbook format. Electrical, thermal, chemical, laser, pressurised-fluid and machine-operation content must be taught and practised only under approved procedures, local risk controls and competent supervision.

Verified source note: Verified sources support process need and classification; USM, abrasive and water-jet processes; EDM/wire EDM; ECM and related processes; and plasma, electron-beam, laser and chemical machining. The four-module organisation is a study sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Advanced Production Process examines modern and non-traditional material-removal methods used where conventional machining is limited by material hardness, fragility, intricate geometry, surface-integrity requirements or specialised production needs. It develops process-selection reasoning grounded in energy source, material compatibility, parameters, quality and safety.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Manufacturing processes, engineering materials, basic electricity, fluid mechanics, workshop drawings and safe machine-shop practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the need for advanced/non-traditional production processes and classify them by material-removal mechanism.
2. Explain abrasive, ultrasonic and jet-based machining processes and their applications.
3. Explain electrical and electrochemical machining processes, their parameters and limitations.
4. Explain thermal and chemical machining processes and compare processes for safe, quality-focused manufacture.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the need, features and classification of advanced production processes.	Understand
CO2	Explain abrasive, ultrasonic and jet-based processes and identify appropriate applications.	Understand
CO3	Explain electrical and electrochemical processes, process variables and work-material restrictions.	Understand
CO4	Explain thermal/chemical processes and select a process using technical, quality and safety criteria.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Need, Classification and Process Selection	Limits of conventional machining; difficult-to-machine materials; energy-based process classification; selection criteria, quality and responsible use.	Approx. 13 hours	CO1	Module 1
2	Mechanical-Energy and Jet Processes	Ultrasonic machining; abrasive-jet machining; water-jet and abrasive-water-jet cutting; finishing concepts, process variables, merits, limitations and applications.	Approx. 16 hours	CO2	Module 2
3	Electrical and Electrochemical Processes	EDM, die-sinking and wire EDM; spark-gap, dielectric and electrode roles; ECM and related electrochemical processes; compatibility, parameters, quality and waste considerations.	Approx. 17 hours	CO3	Module 3
4	Thermal, Chemical and Integrated Process Choice	Plasma-arc, laser-beam and electron-beam machining; chemical machining; surface integrity; process comparison, quality assurance, sustainability and safety.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Need, Classification and Process Selection

Limits of conventional machining; difficult-to-machine materials; energy-based process classification; selection criteria, quality and responsible use.

CO1 · Approx. 13 hours

Mechanical-Energy and Jet Processes

Ultrasonic machining; abrasive-jet machining; water-jet and abrasive-water-jet cutting; finishing concepts, process variables, merits, limitations and applications.

CO2 · Approx. 16 hours

Electrical and Electrochemical Processes

EDM, die-sinking and wire EDM; spark-gap, dielectric and electrode roles; ECM and related electrochemical processes; compatibility, parameters, quality and waste considerations.

CO3 · Approx. 17 hours

Thermal, Chemical and Integrated Process Choice

Plasma-arc, laser-beam and electron-beam machining; chemical machining; surface integrity; process comparison, quality assurance, sustainability and safety.

CO4 · Approx. 14 hours

MODULE 1

Need, Classification and Process Selection

Limits of conventional machining; difficult-to-machine materials; energy-based process classification; selection criteria, quality and responsible use.

Approx. 13 hours **CO1** **Understand · Apply**

Learning goals

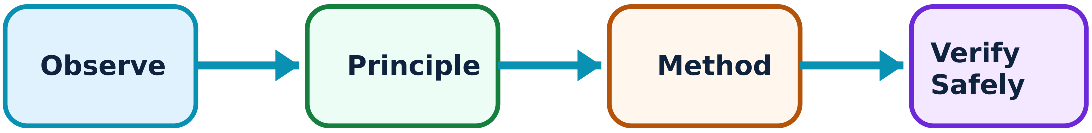
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Need, Classification and Process Selection: identify the system, state its principle, follow the correct method and verify the result safely.

1. Why advanced production processes are used−

Hard, brittle, heat-resistant, electrically conductive or intricate workpieces can make conventional tool-based cutting difficult, uneconomical or damaging. Advanced processes use mechanical, electrical, electrochemical, thermal or chemical energy to remove material in controlled ways.

Study and application method

For a given material and feature, state the conventional-process limitation, required feature/quality and the energy mechanism that could address it.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a given material and feature, state the conventional-process limitation, required feature/quality and the energy mechanism that could address it.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: “Advanced” does not mean universally better. Rate, accuracy, recast layer, heat effect, consumables, operating cost and environmental impact must be compared.

2. Classification and selection logic−

Processes may be grouped as mechanical (USM, AJM, WJM/AWJM), electro-thermal (EDM), electrochemical (ECM), thermal-beam/arc (laser, electron beam, plasma) and chemical. Work-material properties and geometry strongly influence suitability.

Study and application method

Build a selection table with workpiece conductivity, hardness/brittleness, thickness, feature geometry, tolerance, surface requirement, production quantity and safety/environment constraints.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a selection table with workpiece conductivity, hardness/brittleness, thickness, feature geometry, tolerance, surface requirement, production quantity and safety/environment constraints.

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Common mistake / precaution: Never choose a process solely from a textbook capability table; verify current machine capability, guarding, waste handling and qualified process data.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Mechanical-Energy and Jet Processes

Ultrasonic machining; abrasive-jet machining; water-jet and abrasive-water-jet cutting; finishing concepts, process variables, merits, limitations and applications.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

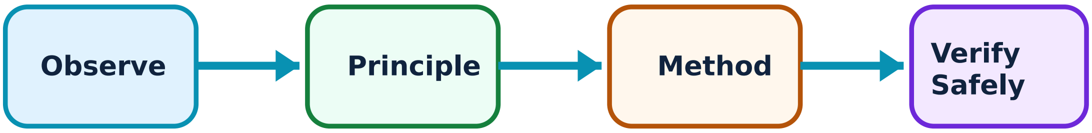
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Mechanical-Energy and Jet Processes: identify the system, state its principle, follow the correct method and verify the result safely.

1. Ultrasonic and abrasive-jet machining−

USM uses high-frequency tool vibration and abrasive slurry to erode suitable hard/brittle materials. AJM uses a high-velocity abrasive stream. Both illustrate mechanical erosion rather than conventional chip formation.

Study and application method

Draw a conceptual USM or AJM layout and identify energy conversion, tool/nozzle, abrasive medium, workpiece, feed/gap and debris-removal path. Relate one parameter to removal rate or finish qualitatively.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a conceptual USM or AJM layout and identify energy conversion, tool/nozzle, abrasive medium, workpiece, feed/gap and debris-removal path. Relate one parameter to removal rate or finish qualitatively.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Abrasive dust, high-frequency equipment, moving components and flying particles require approved enclosures, extraction, eye/face protection and supervised operation.

2. Water-jet, abrasive-water-jet and finishing−

Water-jet processes use a high-velocity fluid stream; abrasive-water-jet adds abrasive for harder materials. Advanced finishing methods can improve surface integrity but have specific material, geometry and rate limits.

Study and application method

Compare WJM and AWJM for a stated material by cutting mechanism, expected kerf/quality, abrasive requirement, likely application and waste stream.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare WJM and AWJM for a stated material by cutting mechanism, expected kerf/quality, abrasive requirement, likely application and waste stream.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Pressurised jets can cause severe injury and equipment damage. Do not use approximate classroom settings as operating parameters; follow the machine-specific safe work procedure.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Electrical and Electrochemical Processes

EDM, die-sinking and wire EDM; spark-gap, dielectric and electrode roles; ECM and related electrochemical processes; compatibility, parameters, quality and waste considerations.

Approx. 17 hours | CO3 | Understand · Apply

Learning goals

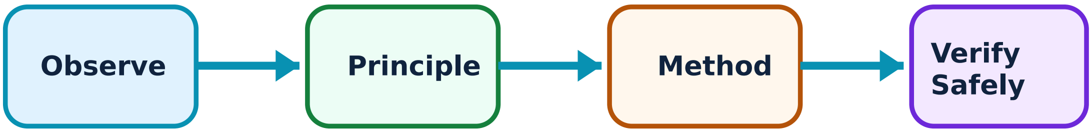
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electrical and Electrochemical Processes: identify the system, state its principle, follow the correct method and verify the result safely.

1. EDM and wire EDM−

EDM removes material through controlled repetitive electrical discharges across a small dielectric gap; both tool and work must be electrically conductive. Wire EDM uses a continuously fed wire to generate programmed contours without a shaped cavity electrode.

Study and application method

Trace the EDM cycle: controlled gap, dielectric breakdown, local melting/vaporisation, pulse termination, flushing and servo control. Then compare die-sinking EDM with wire EDM for a die feature.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace the EDM cycle: controlled gap, dielectric breakdown, local melting/vaporisation, pulse termination, flushing and servo control. Then compare die-sinking EDM with wire EDM for a die feature.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical energy, dielectric fluids, fumes, fire risk and conductive debris demand maintained equipment, approved fluid management, ventilation and emergency controls.

2. ECM and hybrid electrochemical processes−

ECM removes material by controlled anodic dissolution in an electrolyte. It can machine conductive materials without tool-contact cutting force, but electrolyte flow, current, tool shape, gap control and waste treatment are central to quality and responsible practice.

Study and application method

Sketch an ECM cell and identify workpiece polarity, tool, electrolyte circuit, gap and material-removal mechanism. Compare ECM with EDM in thermal effect, tool wear, material restriction and waste.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch an ECM cell and identify workpiece polarity, tool, electrolyte circuit, gap and material-removal mechanism. Compare ECM with EDM in thermal effect, tool wear, material restriction and waste.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Electrolytes, electrical systems and metal-bearing effluent must be handled only through approved containment, PPE and treatment/disposal arrangements.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Thermal, Chemical and Integrated Process Choice

Plasma-arc, laser-beam and electron-beam machining; chemical machining; surface integrity; process comparison, quality assurance, sustainability and safety.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

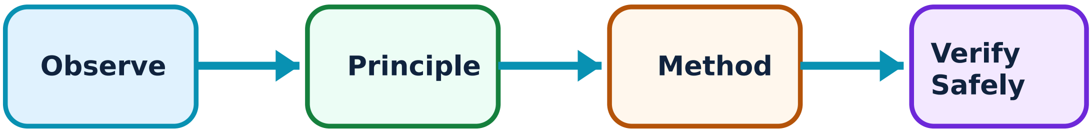
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Thermal, Chemical and Integrated Process Choice: identify the system, state its principle, follow the correct method and verify the result safely.

1. Thermal and beam machining−

Plasma, laser and electron beams concentrate energy to melt, vaporise or otherwise remove material. They can support specialised cutting, drilling or micro-features, while heat-affected zone, taper, recast layer, reflectivity and thickness range require evaluation.

Study and application method

For a proposed feature, compare two thermal processes by energy source, work-material response, likely accuracy/heat effects, throughput and control requirement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a proposed feature, compare two thermal processes by energy source, work-material response, likely accuracy/heat effects, throughput and control requirement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: High-energy beams, plasma arcs, radiation, fume, fire and high voltage require engineered interlocks, authorised enclosures and trained operators—never classroom demonstrations without institutional controls.

2. Chemical machining and final selection−

Chemical machining selectively removes material by controlled chemical action with masking and etching. Across all advanced processes, selection balances technical capability with quality, cost, cycle time, energy, worker protection and waste control.

Study and application method

Complete a documented process-selection justification for a component, including assumptions, alternatives rejected, inspection approach, environmental controls and the limits of the recommendation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Complete a documented process-selection justification for a component, including assumptions, alternatives rejected, inspection approach, environmental controls and the limits of the recommendation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Etchants and process residues require current safety data, compatible materials, spill readiness and regulated waste management; do not improvise chemicals or disposal.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Need, Classification and Process Selection

Use the concept flow to revise observation, principle, method and verification.

Module 2: Mechanical-Energy and Jet Processes

Use the concept flow to revise observation, principle, method and verification.

Module 3: Electrical and Electrochemical Processes

Use the concept flow to revise observation, principle, method and verification.

Module 4: Thermal, Chemical and Integrated Process Choice

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create a process-classification map that links energy type, work-material restrictions and example applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a documented comparison of USM, AJM and AWJM for a brittle ceramic component.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Analyse a simulated EDM/wire-EDM process plan and identify conductivity, flushing, electrode/wire and surface-integrity considerations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a supervised safety-and-environment checklist for a proposed advanced-machining laboratory, including energy isolation, ventilation and waste streams.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Defend a process-selection decision for a complex feature, explicitly stating assumptions and quality-verification needs.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Need, Classification and Process Selection

Explain Why advanced production processes are used and state one correct method, application or precaution.−

Hard, brittle, heat-resistant, electrically conductive or intricate workpieces can make conventional tool-based cutting difficult, uneconomical or damaging. Advanced processes use mechanical, electrical, electrochemical, thermal or chemical energy to remove material in controlled ways.

Answer framework: For a given material and feature, state the conventional-process limitation, required feature/quality and the energy mechanism that could address it.

Important point: “Advanced” does not mean universally better. Rate, accuracy, recast layer, heat effect, consumables, operating cost and environmental impact must be compared.

Explain Classification and selection logic and state one correct method, application or precaution.−

Processes may be grouped as mechanical (USM, AJM, WJM/AWJM), electro-thermal (EDM), electrochemical (ECM), thermal-beam/arc (laser, electron beam, plasma) and chemical. Work-material properties and geometry strongly influence suitability.

Answer framework: Build a selection table with workpiece conductivity, hardness/brittleness, thickness, feature geometry, tolerance, surface requirement, production quantity and safety/environment constraints.

Important point: Never choose a process solely from a textbook capability table; verify current machine capability, guarding, waste handling and qualified process data.

Module 2 — Mechanical-Energy and Jet Processes

Explain Ultrasonic and abrasive-jet machining and state one correct method, application or precaution.−

USM uses high-frequency tool vibration and abrasive slurry to erode suitable hard/brittle materials. AJM uses a high-velocity abrasive stream. Both illustrate mechanical erosion rather than conventional chip formation.

Answer framework: Draw a conceptual USM or AJM layout and identify energy conversion, tool/nozzle, abrasive medium, workpiece, feed/gap and debris-removal path. Relate one parameter to removal rate or finish qualitatively.

Important point: Abrasive dust, high-frequency equipment, moving components and flying particles require approved enclosures, extraction, eye/face protection and supervised operation.

Explain Water-jet, abrasive-water-jet and finishing and state one correct method, application or precaution.−

Water-jet processes use a high-velocity fluid stream; abrasive-water-jet adds abrasive for harder materials. Advanced finishing methods can improve surface integrity but have specific material, geometry and rate limits.

Answer framework: Compare WJM and AWJM for a stated material by cutting mechanism, expected kerf/quality, abrasive requirement, likely application and waste stream.

Important point: Pressurised jets can cause severe injury and equipment damage. Do not use approximate classroom settings as operating parameters; follow the machine-specific safe work procedure.

Module 3 — Electrical and Electrochemical Processes

Explain EDM and wire EDM and state one correct method, application or precaution.−

EDM removes material through controlled repetitive electrical discharges across a small dielectric gap; both tool and work must be electrically conductive. Wire EDM uses a continuously fed wire to generate programmed contours without a shaped cavity electrode.

Answer framework: Trace the EDM cycle: controlled gap, dielectric breakdown, local melting/vaporisation, pulse termination, flushing and servo control. Then compare die-sinking EDM with wire EDM for a die feature.

Important point: Electrical energy, dielectric fluids, fumes, fire risk and conductive debris demand maintained equipment, approved fluid management, ventilation and emergency controls.

Explain ECM and hybrid electrochemical processes and state one correct method, application or precaution.−

ECM removes material by controlled anodic dissolution in an electrolyte. It can machine conductive materials without tool-contact cutting force, but electrolyte flow, current, tool shape, gap control and waste treatment are central to quality and responsible practice.

Answer framework: Sketch an ECM cell and identify workpiece polarity, tool, electrolyte circuit, gap and material-removal mechanism. Compare ECM with EDM in thermal effect, tool wear, material restriction and waste.

Important point: Electrolytes, electrical systems and metal-bearing effluent must be handled only through approved containment, PPE and treatment/disposal arrangements.

Module 4 — Thermal, Chemical and Integrated Process Choice

Explain Thermal and beam machining and state one correct method, application or precaution.−

Plasma, laser and electron beams concentrate energy to melt, vaporise or otherwise remove material. They can support specialised cutting, drilling or micro-features, while heat-affected zone, taper, recast layer, reflectivity and thickness range require evaluation.

Answer framework: For a proposed feature, compare two thermal processes by energy source, work-material response, likely accuracy/heat effects, throughput and control requirement.

Important point: High-energy beams, plasma arcs, radiation, fume, fire and high voltage require engineered interlocks, authorised enclosures and trained operators—never classroom demonstrations without institutional controls.

Explain Chemical machining and final selection and state one correct method, application or precaution.−

Chemical machining selectively removes material by controlled chemical action with masking and etching. Across all advanced processes, selection balances technical capability with quality, cost, cycle time, energy, worker protection and waste control.

Answer framework: Complete a documented process-selection justification for a component, including assumptions, alternatives rejected, inspection approach, environmental controls and the limits of the recommendation.

Important point: Etchants and process residues require current safety data, compatible materials, spill readiness and regulated waste management; do not improvise chemicals or disposal.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Need, Classification and Process Selection.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Mechanical-Energy and Jet Processes.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Electrical and Electrochemical Processes.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Thermal, Chemical and Integrated Process Choice.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Limits of conventional machining; difficult-to-machine materials; energy-based process classification; selection criteria, quality and responsible use..

2. Develop a complete answer or practical plan for: Ultrasonic machining; abrasive-jet machining; water-jet and abrasive-water-jet cutting; finishing concepts, process variables, merits, limitations and applications..
3. Develop a complete answer or practical plan for: EDM, die-sinking and wire EDM; spark-gap, dielectric and electrode roles; ECM and related electrochemical processes; compatibility, parameters, quality and waste considerations..
4. Develop a complete answer or practical plan for: Plasma-arc, laser-beam and electron-beam machining; chemical machining; surface integrity; process comparison, quality assurance, sustainability and safety..

LAST-MILE REVISION

Quick Revision

Module 1: Need, Classification and Process Selection

Limits of conventional machining; difficult-to-machine materials; energy-based process classification; selection criteria, quality and responsible use.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Mechanical-Energy and Jet Processes

Ultrasonic machining; abrasive-jet machining; water-jet and abrasive-water-jet cutting; finishing concepts, process variables, merits, limitations and applications.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Electrical and Electrochemical Processes

EDM, die-sinking and wire EDM; spark-gap, dielectric and electrode roles; ECM and related electrochemical processes; compatibility, parameters, quality and waste considerations.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Thermal, Chemical and Integrated Process Choice

Plasma-arc, laser-beam and electron-beam machining; chemical machining; surface integrity; process comparison, quality assurance, sustainability and safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Why advanced production processes are used	Hard, brittle, heat-resistant, electrically conductive or intricate workpieces can make conventional tool-based cutting difficult, uneconomical or damaging.
Classification and selection logic	Processes may be grouped as mechanical (USM, AJM, WJM/AWJM), electro-thermal (EDM), electrochemical (ECM), thermal-beam/arc (laser, electron beam, plasma) and chemical.
Ultrasonic and abrasive-jet machining	USM uses high-frequency tool vibration and abrasive slurry to erode suitable hard/brittle materials.
Water-jet, abrasive-water-jet and finishing	Water-jet processes use a high-velocity fluid stream; abrasive-water-jet adds abrasive for harder materials.
EDM and wire EDM	EDM removes material through controlled repetitive electrical discharges across a small dielectric gap; both tool and work must be electrically conductive.
ECM and hybrid electrochemical processes	ECM removes material by controlled anodic dissolution in an electrolyte.
Thermal and beam machining	Plasma, laser and electron beams concentrate energy to melt, vaporise or otherwise remove material.
Chemical machining and final selection	Chemical machining selectively removes material by controlled chemical action with masking and etching.

LEARNING RESOURCES

References

Recommended advanced-production-process references

1. V. K. Jain, Advanced Machining Processes.
2. P. C. Pandey and H. S. Shan, Modern Machining Processes.
3. B. Bhattacharyya, Electrochemical Micromachining for Nanofabrication, MEMS and Nanotechnology.
4. M. P. Groover, Fundamentals of Modern Manufacturing.

Approved public advanced-machining sources

- https://www.vedang.in/diploma/study_material/MECH/Advanced_Manufacturing_Process.p
- https://nptel.ac.in/courses/112103202

These sources provide the verified study basis while official Course 5024D detail is unavailable; they do not replace machine manuals, safety data, industrial standards or authorised operating procedures.